# Danielle A. Ripsman

dripsma1@jh.edu | https://danirip.github.io
Google Scholar | github/DaniRip | in/danielleripsman

**Research Summary:** Large-scale and robust optimization for healthcare decisions; learning optimization models and algorithms from data using mixed-integer, Pareto-robust and inverse optimization methods; applications in radiation therapy planning, home care scheduling and the adoption of AI tools in patient-facing care.

## ■ ACADEMIC APPOINTMENTS

**Johns Hopkins University** — *2025 – Present*
**Malone Postdoctoral Fellow, Malone Center for Engineering in Healthcare**
*Whiting School of Engineering, Baltimore, MD, USA*
**Additional Affiliations**: *Center for Systems Science and Engineering (CSSE), Hopkins Business of Health Initiative (HBHI)*
**Mentor**: Dr. Kimia Ghobadi

**University of Toronto** — *2024 – 2025*
**Postdoctoral Fellow in Operations Management & Statistics**
*Rotman School of Management, Toronto, ON, Canada*
**Additional Affiliations**: *Schulich School of Business, York University; Department of Management, University of Toronto Scarborough*
**Mentors**: Dr. Andre Cire and Dr. Adam Diamant

## ■ EDUCATION

**University of Waterloo** — *2018 – 2024*
**Ph.D. in Management Sciences**
*Management Science & Engineering Department, Waterloo, ON, Canada*
**Advisor**: Dr. Houra Mahmoudzadeh
**Thesis**: Large-Scale Optimization Approaches for Radiation Therapy

**University of Waterloo** — *2016 – 2018*
**M.A.Sc. in Management Sciences**
*Management Sciences Department, Waterloo, ON, Canada*
**Advisor**: Dr. Houra Mahmoudzadeh
**Thesis**: Robust Direct Aperture Optimization Methods for Cardiac Sparing in Left-Sided Breast Cancer Radiation Therapy

**University of Toronto** — *2010 – 2016*
**B.A.Sc. in Industrial Engineering**
*Mechanical & Industrial Engineering Department, Toronto, ON, Canada*
Graduated with Honours and an Engineering Business Minor

## ■ PUBLICATIONS

**Refereed Journal Articles**

1. **D. A. Ripsman**, S. H. Hristov, G. Gola, W. Kwong, M. Singh, E. Osei, J. Darko and H. Mahmoudzadeh. "A comparison of voxel sampling approaches for intensity-modulated radiation therapy." *Medical Physics*, 52(12):e70179, 2025. [Link]
2. **D. A. Ripsman**, F. Rahimi, H. Abouee-Mehrizi and H. Mahmoudzadeh. "Light Pareto robust optimization for IMRT treatment planning." *Medical Physics*, 50(5):2637–2648, 2023. [Link]
3. **D. A. Ripsman**, T. G. Purdie, T. C. Y. Chan and H. Mahmoudzadeh. "Robust direct aperture optimization for radiation therapy treatment planning." *INFORMS Journal on Computing*, 34(4):2017–2038, 2022. [Link] [arXiv]
4. **D. A. Ripsman**, D. M. Aleman and K. Ghobadi. "Interactive visual guidance for automated stereotactic radiosurgery treatment planning." *Expert Systems with Applications*, 42(21):8337–8348, 2015. [Link]

**Under Review**

1. F. Parker*, **D. A. Ripsman***, E. Springer and K. Ghobadi. "Learning warm starts for the simplex method from measured solver cost." Under review at the *International Conference on Learning Representations (ICLR)* 2027. arXiv:2610.20892. [Link] (*Co-first authors.)

**Working Papers**

1. F. Rahimi, **D. A. Ripsman**, H. Abouee-Mehrizi and H. Mahmoudzadeh. "Superior light Pareto robust optimization with application to radiation therapy." To be submitted to *Production and Operations Management*.
2. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi. "Early reach and workload associations of an ambient AI scribe in community primary care." To be submitted to *JAMA Network Open*.
3. F. Parker, **D. A. Ripsman**, E. Springer and K. Ghobadi. "Pivoting under partial observability: Exact search, learned policies, and degeneracy in simplex pivot selection." To be submitted to *INFORMS Journal on Computing*.
4. **D. A. Ripsman**, A. Cire and A. Diamant. "A decision diagram approach to multi-period home care scheduling under uncertainty." Manuscript in preparation.
5. **D. A. Ripsman**, S. A. Alev and H. Mahmoudzadeh. "A geometric approach to beam angle selection in radiation therapy treatment planning." To be submitted to *European Journal of Operational Research*.
6. **D. A. Ripsman** and H. Mahmoudzadeh. "Closed-form constraint generation for classes of robust polyhedral uncertainty." Manuscript in preparation.

## ■ PRESENTATIONS

**Conference Presentations**
*All presented by D. A. Ripsman*

1. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload: A multi-site DiD cohort study", INFORMS Healthcare, Raleigh, NC, July 2026.
2. **D. A. Ripsman** and K. Ghobadi "Generative Inverse Optimization for Healthcare Decision Making", INFORMS Healthcare, Raleigh, NC, July 2026.
3. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload", CORS Annual Meeting, Kingston, ON, June 2026.
4. **D. A. Ripsman**, A. Cire and A. Diamant. "Multi-Period Home Care Scheduling Under Uncertainty", CORS Annual Meeting, Edmonton, AB, June 2025.
5. **D. A. Ripsman**, "A Geometrical Approach to Beam Angle Optimization", ICS Conference, Toronto, ON, March 2025.
6. **D. A. Ripsman**, F. Rahimi, H. Abouee-Mehrizi & H. Mahmoudzadeh, "Light Pareto Robust Optimization for IMRT Treatment Planning", CORS Annual Meeting, London, ON, June 2024.
7. **D. A. Ripsman**, "Integrating Uncertainty Mitigation into Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Phoenix, AZ, October 2023.
8. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Healthcare, Toronto, ON, July 2023.
9. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", CORS/Optimization Days, HEC Montréal, QC, May 2023.
10. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Indianapolis, IN, October 2022.

11. **D. A. Ripsman**, H. Mahmoudzadeh, “Structural Solution Approaches to Robust Optimization with Polyhedral Uncertainty”, CORS/INFORMS International, Vancouver, BC, June 2022.
12. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, “Closed-form Constraint Generation for Classes of Robust Polyhedral Uncertainty”, INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.
13. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, “A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning”, INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.
14. **D. A. Ripsman**, H. Mahmoudzadeh, “Closed-form constraint generation for robust radiation therapy treatment planning”, CORS Annual Meeting, (Virtual), May 2021.
15. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning”, INFORMS Healthcare, Cambridge, MA, July 2019.
16. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning”, Canadian Healthcare Optimization Workshop, Saskatoon, SK, May 2019.
17. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning”, CORS Annual Conference, Halifax, NS, June 2018.
18. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, “Robust Mixed Integer Optimization for Radiation Therapy Treatment Planning with Delivery Constraints”, Optimization Days, Montreal, QC, May 2017.
19. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. “Interactive visual decision support for radiation therapy treatment planning”, Industrial & Systems Engineering Research Conference (ISERC), Montreal, QC, June 2014.
20. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. “Interactive visual decision support for radiation therapy treatment planning”, Undergraduate Engineering Research Day (UnERD), University of Toronto, ON, August 2013.
    **Second place in the Industrial Engineering Oral Presentations category.**

**Poster Presentations**
*All presented by D. A. Ripsman*

1. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. “The Impact of AI Scribe on Primary Care Workload”, Department of Medicine and Whiting School of Engineering Research Retreat, Baltimore, USA, 2026.
2. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. “The Impact of AI Scribe on Primary Care Workload”, Johns Hopkins Research Symposium on Engineering in Healthcare, Baltimore, USA, 2025.
3. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. “Graphical Interface for Interactive Parameter Selection in Radiation Therapy Treatment Planning”, MIE Research Symposium, University of Toronto, Toronto, Canada, June 2013.

## ■ FUNDING

### Fellowships & Scholarships

- *2025 – Present* ▪ Malone Postdoctoral Fellowship, Malone Center for Engineering in Healthcare, Johns Hopkins University
- *2019 – 2022* ▪ NSERC Canada Graduate Scholarship–Doctoral (CGS-D) ($105,000; $35,000 for 3 years)
- *2018 – 2022* ▪ Engineering Excellence PhD Fellowship ($120,000; $30,000 for 4 years)
- *2018* ▪ Provost’s Doctoral Entrance Award for Women ($5,000)
- *2017* ▪ University of Waterloo Graduate Scholarship ($750)

*2013* ▪ NSERC Undergraduate Student Research Award ($4,500)

*2010 – 2011* ▪ John Hirschorn Memorial Scholarship ($20,000; $10,000 for 2 years)

*2010* ▪ University of Toronto Scholars Scholarship ($5,000)

## AWARDS & HONOURS

### Teaching Awards

*2020, 2022* ▪ Sandford Fleming Foundation Teaching Assistant Award, for the R laboratory of the introductory statistics course, University of Waterloo

### Competitions & Honours

*2024* ▪ **Finalist** (Health Care Operations Research SIG and Open categories), CORS Student Paper Competition, London, ON, for "Superior and light Pareto robust optimization"

*2019* ▪ **Winner**, CORS Health Care Operational Research SIG Student Presentation Competition, Saskatoon, SK

*2018* ▪ **Finalist**, Three Minute Thesis (3MT), University of Waterloo (first place in the departmental heat)

*2014* ▪ Third place, Simulation Competition, IIE National Student Conference, Toronto, ON

*2013* ▪ Second place, Industrial Engineering Oral Presentations, Undergraduate Engineering Research Day, University of Toronto, ON

*2013* ▪ Third place, Operations Research Case Competition, Industrial Engineering Competitions Day, University of Toronto, ON

*2012* ▪ First place, Student Design Competition, American Society of Mechanical Engineers, University of Toronto, ON

## GRANT WRITING EXPERIENCE

**NIH R01 (NHLBI)**
*2026 – 2030*

**SCH: Intelligent Decision-Support System for ICU Respiratory Management**
*R01HL192731, NSF–NIH Smart and Connected Health program. PIs: Dr. James Fackler, Dr. Kimia Ghobadi and Dr. Chien-Ming Huang.*
Contributed to proposal development and writing.

## TEACHING EXPERIENCE

**Course Instructor**
*May – Aug 2023*

**MSCI 331: Introduction to Optimization, University of Waterloo**
*Sole instructor of an evening section; 112 upper-year undergraduates*

- **Student Course Perceptions: 4.29/5 average across the four instructor characteristics**, above the Faculty of Engineering average of 3.98 (item range 4.04–4.55; 29 of 112 responding)
- Designed and administered online auto-graded assignments, written quizzes, midterms and final examinations; supervised two teaching assistants
- Held full administrative responsibility for the section, including announcements, grade entry and documentation

**Guest Lecturer**
*Fall 2026*

**EN.560.450/650: Introduction to Operations Research**
*Johns Hopkins University, in Dr. K. Ghobadi's course*

**Teaching Assistant** — **Dept. of Management Science & Engineering, University of Waterloo**
*Twelve terms across eight courses, 2017–2023*

- *2019 – 2023* — *MSCI 251: Probability & Statistics for Engineers I*. Designed and taught the applied R laboratories, and rebuilt them for remote delivery during the pandemic (5 terms; Dr. H. Mahmoudzadeh)
- *2023* — *MSCI 734: Network Models*. Assignment and project grading (Dr. F. Gzara)
- *2022* — *MSCI 435: Advanced Optimization Techniques*. Tutorials, grading and student support (Dr. S. Elhedhli)
- *2022* — *MSCI 703: Applied Optimization*. Grading and solution manuals (Dr. S. Elhedhli)
- *2019* — *MSCI 401/402: Design Project*. Course logistics and discussion sessions (Dr. A. Hurst)
- *2018* — *Work Report Marking*. Assessment and written feedback on work-term reports (D. Chan)
- *2017* — *MSCI 436: Decision Support Systems*. Built and taught weekly Visual Basic tutorials, office hours, grading; 42 students (Dr. M. Karimi)
- *2017* — *MSCI 331: Introduction to Optimization*. Tutorials, office hours, grading and posted solutions; 41 students (Dr. A. Hiassat)

**Courses Prepared to Teach**

- *Taught:* introduction to optimization, and linear and integer programming (sole instructor); probability and statistics for engineers (5 terms of R laboratories)
- *Foundations:* introductory programming (C++, Java, VB, MATLAB, Python); applied linear algebra
- *Advanced:* optimization under uncertainty (robust and stochastic); large-scale optimization; simulation
- *Applied:* healthcare systems engineering and analytics; decision support systems

## ■ RESEARCH MENTORSHIP

**Undergraduate Researchers** — **Dept. of Management Science & Engineering, University of Waterloo**
*Six undergraduate co-op students and research assistants supervised, 2018–2023*

- *2023* — **Gatik Gola** (co-op): motivated the project, recommended methods and ran weekly meetings for a CT voxel sampling study; *co-author, Medical Physics 2025*
- *2023* — **Karina Wilk** (co-op): onboarding and weekly planning for the hospital optimization pipeline begun during the Grand River Hospital internship
- *2021 – 2022* — **William Kwong** (research assistant): proposed and oversaw an interface for running sampling protocols for fluence map optimization; *co-author, Medical Physics 2025*
- *2021* — **Tiantian Li** (research assistant): exploratory project adapting open-source radiation therapy tools (CERR, matRad) to optimization software
- *2019* — **Gogilan Selvarajah** (research assistant): pair programming and project direction for a sliding-window IMRT visualization
- *2018* — **Jessie Yeung** (research assistant): background research and data acquisition for an IMRT sliding-window modelling project

## RESEARCH & CLINICAL EXPERIENCE

**Graduate Researcher** — *Sept 2016 – Aug 2024*
**Dept. of Management Science & Engineering, University of Waterloo**
*Supervised by Dr. Houra Mahmoudzadeh.*
- Built optimization models for radiation therapy treatment planning, tailored to breathing motion uncertainty in breast cancer treatment
- Prototyped and tested models against large clinical datasets using a custom C++/CPLEX code base
- Implemented structured visualizations of plan quality in MATLAB

**Clinical Research Intern** — *Oct – Dec 2018*
**Medical Physics Department, Grand River Hospital, Kitchener, ON**
*Supervised by Dr. Ernest Osei.*
- Built the I/O bridge between Varian Eclipse and open-source CERR modules for importing clinical DICOM plans
- Wrote custom collimator rotation code and validated the pipeline on patient plans

**Thesis Student** — *Sept 2015 – Apr 2016*
**Toronto Intelligent Decision Engineering Laboratory, University of Toronto**
*Supervised by Dr. J. Christopher Beck.*
- Developed A*-based batch scheduling algorithms, with C++ implementations of Dijkstra and A*

**Capstone Student** — *Sept 2015 – Apr 2016*
**Addiction Medicine Services, Centre for Addiction and Mental Health, Toronto, ON**
*Supervised by Dr. Michael Carter.*
- Streamlined clinic scheduling and resource allocation, and built a simulation to validate the recommended interventions

**Undergraduate Researcher** — *Fall 2012 – May 2014*
**Medical Operations Research Laboratory, University of Toronto**
*Supervised by Dr. Dionne Aleman.*
- Took two interactive radiosurgery planning prototypes from concept to completion in self-taught MATLAB

**Case Study Author** — *Fall 2014*
**Industrial Engineering Department, University of Toronto**
*Case adopted as a teaching aid in Cases in Operational Research (MIE367).*
- Wrote a multi-criteria decision-making case around a restaurant venture, with realistic data distributions and a financial planning simulation

## PROFESSIONAL SERVICE

**Journal Refereeing**
Referee for *Computers & Operations Research*, *European Journal of Operational Research*, *International Transactions in Operational Research*, *Medical Physics*, *Operations Research Forum* and *Quantitative Imaging in Medicine and Surgery*

**Conference Service**
- Session Chair, INFORMS Computing Society (ICS) Conference, Toronto, ON, 2025; also a conference volunteer
- Session Chair, CORS/INFORMS International Conference, Vancouver, BC, 2022
- Designed the virtual conference venue (Gather Town) for the CORS Annual Conference, 2021, with Dr. Kiefer Burgess

**Memberships**
Institute for Operations Research and the Management Sciences (INFORMS), including its Health Applications Society (HAS); Canadian Operational Research Society (CORS)

## LEADERSHIP & OUTREACH

**National Student Councillor** — *2024 – 2025*
**Canadian Operational Research Society (CORS)**
*Elected to represent the national student body.*
- Led the post-pandemic revival of student chapters across Canada

**Invited Session Leader** — *May 2024*
**SPARCS Workshop (CEMC), University of Waterloo**
*Week-long computer science workshop for Grade 9 and 10 students of gender identities underrepresented in computing.*
- Adapted the TORCH format into a two-hour introduction to operations research for 40 students: ten teams competed on two optimization puzzles, followed by a talk linking their strategies to real-world problems

**President** — *Sept 2017 – 2020*
**The Operations Research Challenge (TORCH), University of Waterloo**
*Founded the Waterloo-region edition of the annual high school competition.*
- Led logistics for the 2018, 2019 and *2020*[*] competitions: student and volunteer recruitment, funding, testing and scheduling
- Wrote competition problems and validators, and coordinated the automated grading software, with volunteers from Waterloo, Concordia and Toronto

**Student Chapter President** — *Sept 2017 – Sept 2018*
**Canadian Operational Research Society (CORS), Waterloo, ON**
*Ran the local student chapter's speaker and social programme.*
- Coordinated speakers and events, onboarding, communications and website oversight
- Grew active membership from fewer than 10 to nearly 30

**President** — *Sept 2016 – Aug 2017*
**Management Sciences Student Association, University of Waterloo**
*Elected departmental representative for graduate students.*
- Budgeted and ran invited speaker and social events; represented students at monthly departmental and cross-departmental meetings

**Executive Member** — *Fall 2014 – Winter 2016*
**The Operations Research Challenge (TORCH), University of Toronto**
*Co-Chief Creative Officer (2016); Assistant Editor-in-Chief (2015).*
- Oversaw creation of the 2016 contest, edited the 2015 problem set, and wrote a scheduling problem with an automated solver for grading

## INDUSTRY EXPERIENCE

**Rational Software Developer** — *May 2014 – Aug 2015*
**IBM Canada Software Lab, Markham, ON**
*Eclipse-based enterprise software development, testing and debugging (TPF Toolkit, RDi); supervised by Rob Cecco.*

**Engineering Intern** — *Summer 2012*
**Rotork Controls Canada, Mississauga, ON**
*Engineering drawings, customer documentation, and a Visual Basic database for intracompany drawing retrieval.*

## TECHNICAL SKILLS

**Languages & Tools** — Python, C++, Java, MATLAB, R, LaTeX

**Optimization** — CPLEX, Gurobi, HiGHS, AMPL, Excel Solver

**Clinical / Domain** — Varian Eclipse, CERR, matRad, DICOM-RT

[*]Cancelled due to the pandemic